

# Rocket & Flight Computer

High-power rocket built with custom avionics, 3D-printed aerostructures, parachute recovery, key-locked ignition.

LUKE SCHREIBER & TEAM

EMBEDDED C++ / ESP32-S3 / SOLIDWORKS / OPENROCKET

**FLIGHT COMPUTER** ESP32-S3, hand-soldered perfboard, SD logging

**SENSING** BME280 barometric altitude

**POWER** TPS61023 boost converter

**STRUCTURES** 3D-printed fins, nose cone, motor mount

**RECOVERY** Parachute, validated canopy geometry

**IGNITION** Custom igniter, key-locked interlock



Final assembled rocket

## BUILD PROCESS

**01**  
**DESIGN**  
SOLIDWORKS

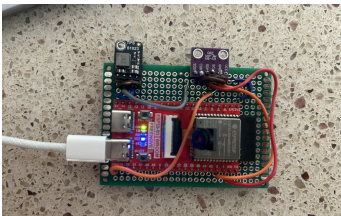
**02**  
**SIMULATE**  
OpenRocket

**03**  
**FABRICATE**  
3D print & assemble

**04**  
**AVIONICS**  
Solder & code

**05**  
**INTEGRATE**  
Bench test

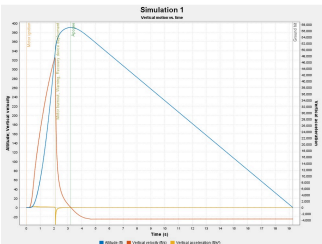
**06**  
**FLY**  
Field launches



**Flight Computer**  
ESP32-S3 + BME280



**Fin & Motor Mount**  
3D printed



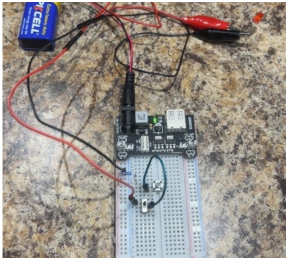
**Flight Simulation**  
OpenRocket



**Launch Setup**  
Field test

## ENGINEERING HIGHLIGHTS

**Igniter built from scratch**  
No off-the-shelf igniter



**Safety Lock**  
Keyed interlock prototype

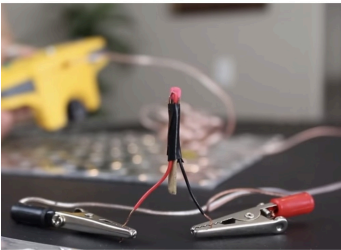
**Key-locked interlock**  
Blocks accidental ignition



**Recovery Parachute**  
Canopy layout

**Validated recovery**  
Canopy sized for safe descent

**Full integration**  
One flight-ready stack



**Custom Igniter**  
Bench test